13. Lorentz Time Signal Composition & Auto-Scaling Math

Time Dilation DAW Technical Proof Series (v0.0.1)

1. Lorentz Velocity Addition Theorem

Time signals in time.math~, time.scope, and time.xy process relativistic dilation vectors gamma.

Combining two time sources with factors gamma_1 and gamma_2 follows Einstein velocity composition with speed-of-light c = 4.0:

$$v_1 = \gamma_1 - 1.0, \quad v_2 = \gamma_2 - 1.0, \quad v_{comp} = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}$$

$$\gamma_{comp} = 1.0 + v_{comp} = 1.0 + \frac{\gamma_1 + \gamma_2 - 2.0}{1.0 + \frac{(\gamma_1 - 1.0)(\gamma_2 - 1.0)}{c^2}}$$

2. Dynamic Auto-Scaling & Phase Orbit Radius

For continuous time telemetry in time.scope, max peak M = max |y[i]| auto scales headroom dynamically:

$$S[n] = S[n-1] + \lambda \cdot (\max(1.0, M \cdot 1.15) - S[n-1])$$

For dual-time 2D Lissajous phase orbits in time.xy, max radial extent R_{max} = max sqrt(x^2 + y^2) auto-scales 2D bounds:

$$R_{max}[n] = R_{max}[n-1] + \lambda \cdot (\max(1.0, R_{max} \cdot 1.15) - R_{max}[n-1])$$